

Book 11: Breeder Flock Management & Hatchery Operations

A Comprehensive Guide to Parent Stock Management, Incubation Science & Day-Old Chick Quality

PoultryWise Operational Manuals • Breeding & Incubation Management Standard

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CHAPTER 1: PARENT STOCK (PS) REARING & BODY WEIGHT CONTROL

Managing parent stock requires strict feed restriction and weight monitoring to ensure uniform flock sexual maturity and high peak hatchability.

CHAPTER 2: MALE-TO-FEMALE RATIOS & MATING DYNAMICS

Optimizing male fertility involves managing male-to-female ratios (typically 9–10 males per 100 females) and spiked male introductions to maintain mating activity throughout the lay cycle.

CHAPTER 3: HATCHING EGG SELECTION, CARE & STORAGE

Proper egg handling prior to incubation directly influences embryonic viability and hatch rates.

Storage Period	Target Temperature (°C)	Target Relative Humidity (%)	Expected Hatchability Impact
1 – 3 Days	18 – 20°C	75 – 80%	Optimal (Baseline 100%)
4 – 7 Days	15 – 17°C	80 – 85%	Negligible reduction (<0.5%/day)
8 – 14 Days	12 – 14°C	80 – 85%	Moderate decline (Requires PREG/SPIDES)

CHAPTER 4: HATCHERY SANITATION & BIOSECURITY PROTOCOLS

Strict one-way traffic flow and automated chemical fogging prevent horizontal pathogen transmission (e.g., *Aspergillus*, *Salmonella*) inside the hatchery facility.

CHAPTER 5: PRE-INCUBATION & EGG SANITATION METHODS

Sanitizing egg shells via thermal fogging or ozone before pre-heating prevents bacterial penetration into the egg pores.

CHAPTER 6: INCUBATION PARAMETERS: TEMPERATURE, HUMIDITY & TURNING

Incubator conditions during single-stage setting require multi-point control to maximize chick yield.

Core Single-Stage Incubation Target

Maintain embryonic shell temperature at 37.8°C (100.0°F), relative humidity between 50–55%, and egg turning at a 45° angle once every hour for the first 18 days.

CHAPTER 7: EMBRYONIC DEVELOPMENT & CANDLING ANALYSIS

Candling eggs at day 7 and day 18 identifies clear (infertile) eggs and early/late embryonic mortality, allowing precise hatchability audits.

CHAPTER 8: HATCHING PHASE & DAY-OLD CHICK PROCESSING

Transferring eggs to hatchers at day 18, managing narrow hatch windows, automated chick separators, feather sexing, and subcutaneous vaccination.

CHAPTER 9: CHICK QUALITY ASSESSMENT & PASGAR SCORE

Evaluating chick vitality using the Pasgar Score (checking navel condition, legs, reflex response, and yolk sac absorption) ensures high field performance.

CHAPTER 10: HATCHERY WASTE MANAGEMENT & LOGISTICS

Environmentally sound disposal of eggshells and unhatched eggs alongside climate-controlled transport vehicles for day-old chick delivery.